BIG DATA: Fundamentals and Applications

Specialization in Data Science

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Faculty of Economics

News

- https://huggingface.co/papers/date/2026-05-14
- https://www.youtube.com/watch?v=OMhKgQmeMhI&t=6s
- https://www.databricks.com/dataaisummit
- https://arxiv.org/
- https://www.linkedin.com/feed/

Project

Figure: Project

Project

Some Examples

- **https://github.com/yeiscop/Sewers-Rover**
- **https://github.com/ANDRES01102023/especializacionbigdata/tree/desarrollo**
- **https://github.com/jp0617/modelo_predictivo**
  https://modelo-predictivo-picr.onrender.com/predict
- **https://github.com/daniaceal-art/bigdata-scoring**
- **https://github.com/danielg7v/proyecto-iam-medallion**
- **https://github.com/natyhiberia/ForestLake**
- **https://github.com/Linaisa96/TrabajoFinalBigData/tree/main**
- **https://github.com/CoRttIs07/BigData—Modelo-churn**

Contents

What is Big Data?

Big Data involves the management, processing, and analysis of massive amounts of data, which cannot be handled using conventional methods and analysis of large-scale datasets that exceed the capabilities of traditional data processing systems.

Figure: 5 V's of Big Data

What is Big Data?

This discipline has become essential in today's world, where information is generated rapidly and in a wide variety of formats. The main goal is to extract value from this data, enabling organizations to better understand their environment, anticipate trends, and make informed decisions.

These datasets are generated at high speed and in multiple formats (structured, semi-structured, and unstructured). The primary objective of Big Data is to extract value from data, enabling organizations to:

- Understand patterns
- Anticipate trends
- Make data-driven decisions

Course Purpose

The purpose of the course is for students to acquire a deep understanding of the technologies, techniques and workflows associated with Big Data.

Figure: Big Data Innovation Hub

Methods for handling large volumes of information are covered, along with strategies that facilitate efficient data processing and analysis.

Course Purpose

Rather than focusing on theory, the course emphasizes:

- Real-world applications
- Hands-on data processing
- End-to-end solution building

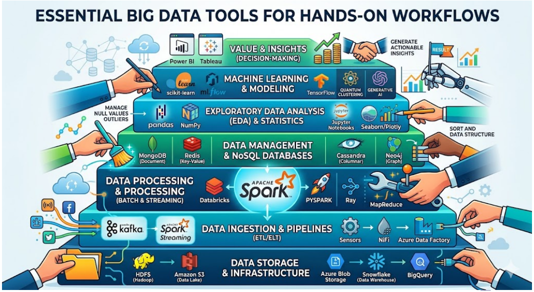
Students will learn how to design, process, analyze, and model data using **modern tools** in a limited time frame.

Figure: capturing and processing live sensor data

Specialized Tools and Platforms

Throughout the course, various specialized **tools** and **platforms** in Big Data are introduced. These technological solutions allow relevant information to be extracted from large data sets, facilitating informed decision-making in complex business scenarios. Knowledge and proficiency in these **tools** are essential to harness the full potential of Big Data.

Figure: Essential **tools** for a hands-on Big Data workflow



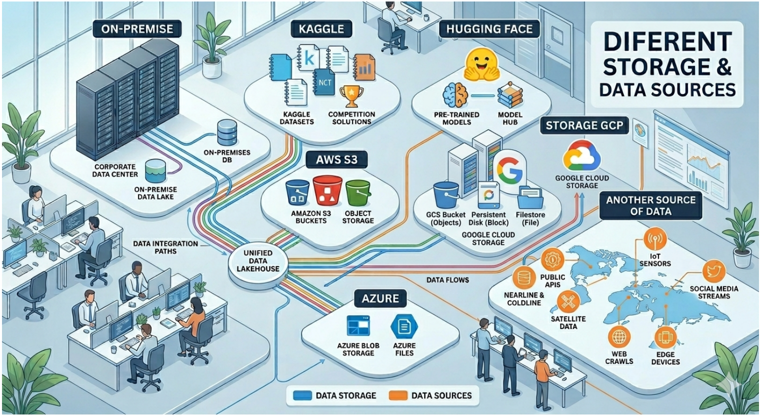
Specialized Tools and Platforms

Figure: Essential **Platforms** for our course

Source: Gemini

Specialized Tools and Platforms

Figure: Essential **storage** for our course



Source: Gemini

Specialized Tools and Platforms

The Cloud Ecosystem for Big Data:

Infrastructure Providers

- **AWS:** Scalability & Maturity.
- **Azure:** Enterprise Integration (Power BI and Microsoft enviroment).
- **GCP:** Specialized AI & BigQuery.

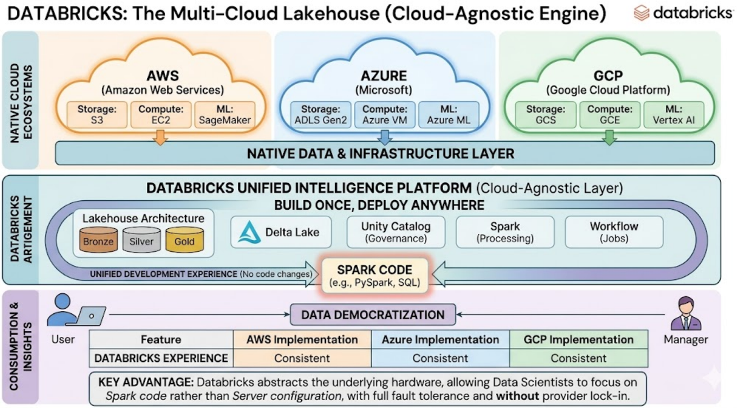
The Databricks Advantage

Databricks provides a **Unified Interface** across all three.
It abstracts the underlying hardware, allowing Data Scientists to focus on **Spark code** rather than **Server configuration**.

Specialized Tools and Platforms

The Cloud Ecosystem for Big Data:

Figure: The Multi-Cloud Lakehouse



Specialized Tools and Platforms

Exploring AWS for Data Science

- **Storage (S3):** The "Infinite" hard drive for Data Lakes.
- **Processing (EMR/Glue):** The engines that transform raw data into insights.
- **Machine Learning (SageMaker):** A full environment for training AI models.

Student Activity Idea

Explore **SageMaker** (ML), **S3**, **EC2**.

Key Lesson: *Cloud makes Big Data "Elastic"—you only pay for the servers while you are using them.*

https://aws.amazon.com/es/training/awsacademy/

Specialized Tools and Platforms

Exploring Google Cloud (GCP) for Data Science

- **BigQuery:** A serverless data warehouse that separates storage from compute. Highly optimized for SQL.
- **Vertex AI:** A unified platform for the entire ML lifecycle (AutoML, training, and deployment).
- **Cloud Storage (GCS):** High-performance, regional or multi-regional object storage.

Why GCP?

GCP provides the most mature environment for **Vertex AI Pipelines**, which would be ideal for orchestrating your experiments and tracking model versions.

https://cloud.google.com/free

Practice: Create Project in GCP wit BILL **U$**

Phase 1: Ingestion to BigQuery

- Open the **BigQuery Console** within your active GCP Project.
- Click the three vertical dots next to your dataset (`gcp_class_practice`) → Select **Create Table**.
- Set Source to **Upload**, choose your local `datos_taxi.csv`, and check the **Auto detect Schema** option.

Phase 2: Managed Training via Vertex AI

- Navigate to **Vertex AI** → Enable recommended APIs → Click **Datasets** → **Create**.
- Select **Tabular Classification**, then select the radio button to link data directly from your BigQuery table path.

Specialized Tools and Platforms

Exploring Google Cloud (GCP) for Data Science

Practice: Create Project in GCP with BILL **U$**

- Click **Train New Model**. Choose **AutoML Classification**, select `high_tip` as the target, and set a budget of 1 node-hour.

Phase 3: Deployment & Infrastructure Teardown

- Once status changes to *Finished*, evaluate model metrics (Confusion Matrix, Feature Importance).
- Go to the **Deploy & Test** tab and deploy the model artifact to an active **Online Endpoint**.
- Input test features into the UI to verify real-time prediction outputs via JSON payloads.
- **CRITICAL COST CONSTRAINT:** Immediately after verifying the API output, you must click **Undeploy model from endpoint** to terminate ongoing cloud compute resource charges.

Specialized Tools and Platforms

Azure: The Enterprise Big Data Ecosystem

- **Azure Databricks:** A first-party service for high-performance Spark clusters and collaborative Data Science.
- **ADLS Gen2:** The underlying storage layer that enables the Medallion Architecture (Bronze/Silver/Gold).
- **Azure ML Studio:** A unified environment for building models using Code, Low-Code (Designer), or No-Code (AutoML).

https://azure.microsoft.com/en-us/free/students

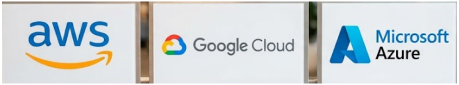
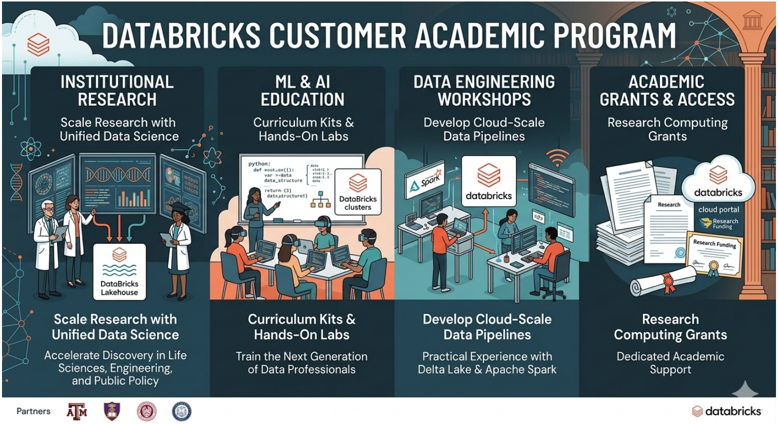


Figure: Caption

Specialized Tools and Platforms

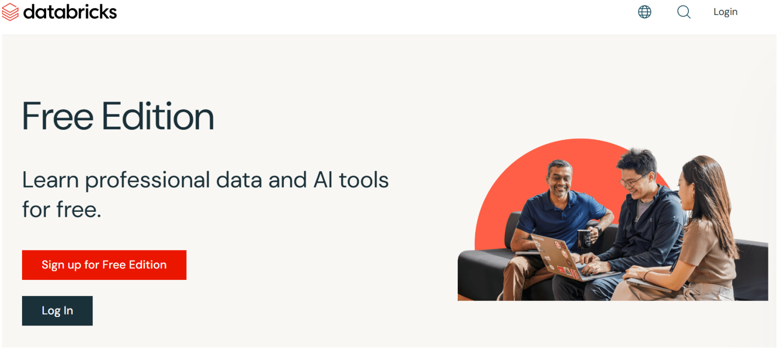
DataBricks Customer Academic



- *https://customer-academy.databricks.com/pages/126/customer-academy-home-page*

Specialized Tools and Platforms

DataBricks Platform



- *https://www.databricks.com/learn/free-edition*

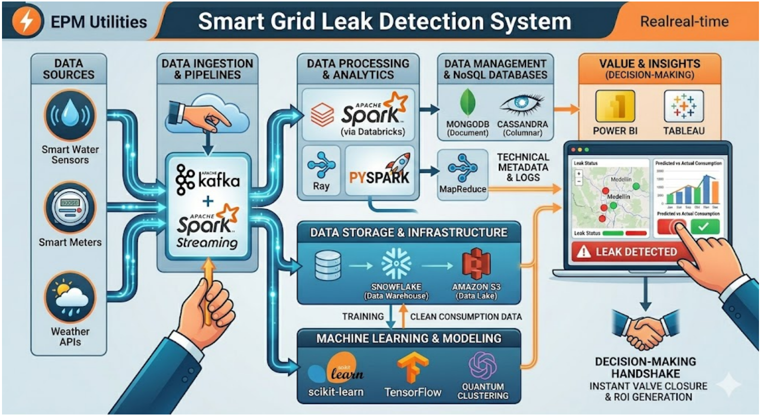
Implementations & Applications

An example of a **water utility company** *trying to detect potential leaks or customer fraud based on service pressure.*

Phase	Tool (Pyramid)	Real-World Action (Utility)
Capture	Kafka / Sensors	Receive 10,000 pulses per second
Compute	Spark / Databricks	Is the pressure $<$ 40 PSI?
Store	Snowflake / S3	Save for the 10-year historical record
Analyze	Scikit-learn / Ray	Find clusters of high-waste users
Display	Power BI	Show the CEO the ROI of the system

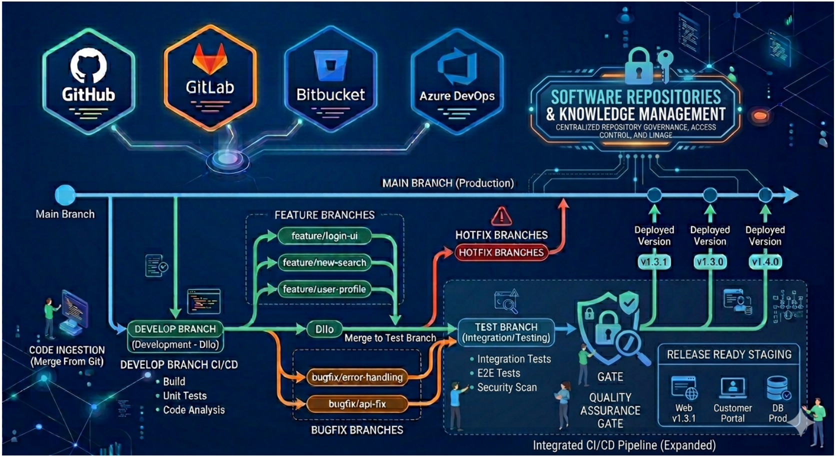
Table: End-to-End Solution Lifecycle - Architecture

Figure: Example Architecture for utilities company



Implementations & Applications

Figure: Repositories - **Know How of CODE**



1. Account Registration & Identity

- Navigate to `github.com` → Click **Sign Up**.
- **Identity Rule:** Select a professional, recognizable username handle (e.g., `firstname-lastname` or institutional ID).

2. Metadata Customization

- Access **Your Profile** → Click **Edit Profile**.
- Populate fields: Full Name, Professional Bio, Organization, and links to your professional research sites.

3. Fork

→ https://github.com/yeiscop/Big_DATA
→ Linked accounts

Practice_1

Leak Water
Example Financial Sector

Utilizing advanced remote water consumption metering, our system detects anomalies and potential water service fraud for utility operations.

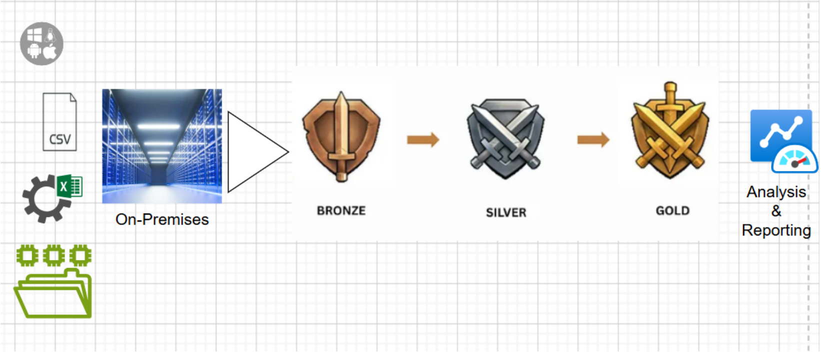


Figure: Example On-Premises Storage

Data Source: 32,581 records of loan applications.
https://www.kaggle.com/datasets/laotse/credit-risk-dataset/data

1. **Bronze Layer:** Ingestion of `credit_risk_dataset.csv`. No modifications to preserve the original audit trail.
2. **Silver Layer:**
   - Imputation of missing *Interest Rates*.
   - Filtering outliers (Age > 100).
   - Standardization of *Employment Length*.
3. **Gold Layer:** Generation of the **Risk Segment** (Critical/Warning/Safe) for Credit Committee consumption.

Unity Catalog Value

Allows us to track the **Lineage** from the raw Kaggle CSV to the final decision, ensuring the bank is compliant with "Right to Explanation" laws.

Figure: Scoring: Cloud Storage or Source

Implementation of Big Data Solutions in Organizations

Organizations can apply **Big Data** solutions to improve decision-making, optimize processes, and gain competitive advantages. When implementing these solutions, they must consider the **ethical** and **security** implications that arise from handling large volumes of data.

Key aspects include protection against **cyberattacks**, the use of sensors for data collection, and responsible management of customer information.

Learning Outcomes

By the end of the course, students will be equipped to develop, implement, and optimize **Big Data** solutions.

They will use advanced tools and techniques for processing and analyzing **large amounts** of data, ensuring that decisions are supported by accurate and relevant information.

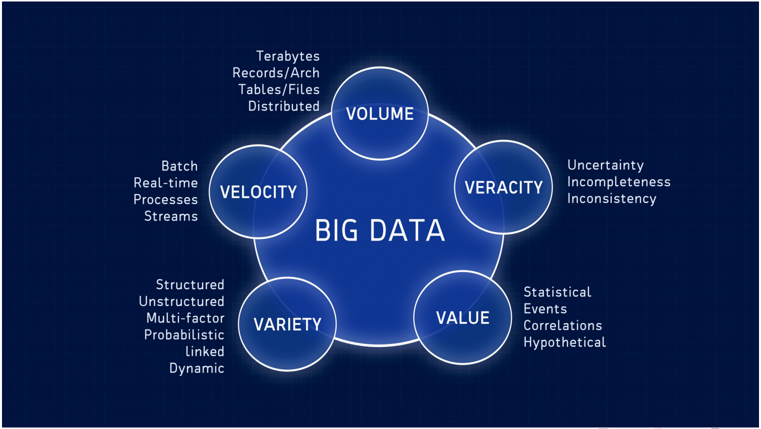
Additionally, the topic of **Small Data** will be mentioned, recognizing its importance in contexts where managing smaller-scale data can be equally significant for decision-making.

Understand the fundamentals and architecture of Big Data systems

- Traditional Data vs Big Data
- The 5 V's of Big Data (Volume, Velocity, Variety, Veracity, Value).
- Architecture: Introduction to Hadoop (HDFS) and the concept of distributed computing.
- Apache Spark (why it is the industry standard)
- Cloud vs. On-Premise: Brief overview of AWS, Azure, Google Cloud and Databricks as a unified analytics platform.

Introduction to Big Data & Ecosystems
The 5 V's

Figure: The 5 V's: A Real-World



Volume The sheer scale of data (e.g., millions of banking transactions).

Velocity The speed of generation (e.g., real-time sensor streams).

Variety Different formats (CSV, TXT, JSON, Images).

Veracity The "truth" or quality of data (Cleaning dirty Kaggle sets).

Value The ultimate goal: Turning bits into business insights.

Traditional Data

- *Structure:* Relational (SQL).
- *Volume:* Gigabytes/Terabytes.
- *Scaling:* Vertical (Bigger Servers).

Big Data

- *Structure:* Unstructured/Semi-structured.
- *Volume:* Petabytes/Exabytes.
- *Scaling:* Horizontal (Clusters).

The Academic Definition

Big Data is not just "large" data; it is data that exceeds the processing capacity of conventional database systems.

- **Hadoop (HDFS):** The pioneer. Introduced "Data Locality"—moving the code to the data, not the data to the code.
- **Apache Spark:** The Engine. Replaced Hadoop's slow disk-based MapReduce with **In-Memory** processing (100x faster).
- **Databricks:** The Platform. A unified environment that integrates Spark, Unity Catalog, and Collaborative Notebooks.

Academic Use Case

We use **Databricks Community Edition** to simulate a multi-node cluster environment without the cost of physical hardware.

2006: Hadoop (HDFS)
Focus: Cheap Storage / Disk-based

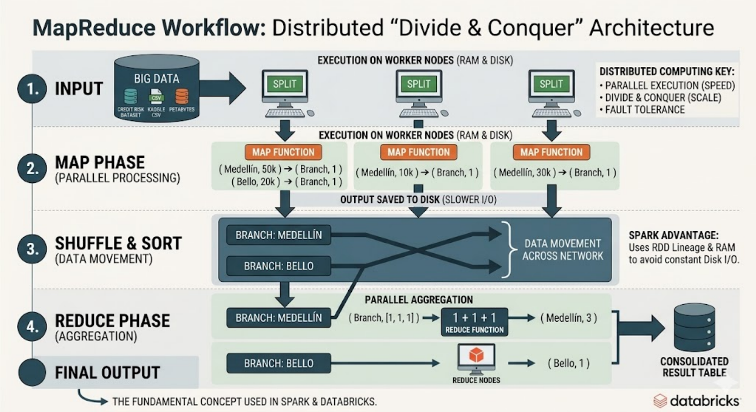
↓

2012: Apache Spark
Focus: In-Memory Speed / Analytics

↓

2015+: Databricks
Focus: Lakehouse / Governance / AI

Figure: RDDs (Resilient Distributed Datasets)



Traditional Server

- Single Machine.
- *Vertical Scaling.*
- Vulnerable to crashes.
- Good for small SQL DBs.

Big Data Cluster

- Group of Nodes.
- *Horizontal Scaling.*
- High Availability.
- Required for **Spark**.

The Spark Advantage

Spark splits a 1GB file into 10 pieces, sending one piece to each Node. They process the data 10x faster than a single server would.

Unit 1 Checkpoint: Big Data Fundamentals

1. **Architectural Scaling:** To handle a massive increase in data volume, is it better to perform *Vertical* or *Horizontal* scaling in a Big Data ecosystem?
2. **The 5 V's:** Which "V" refers to the quality and trustworthiness of data (e.g., handling "dirty" values in the Kaggle dataset)?
3. **Engine Evolution:** Why does Apache Spark outperform the traditional Hadoop MapReduce model?
4. **Medallion Layers:** Which layer (*Bronze, Silver, or Gold*) acts as the immutable "Source of Truth" for raw CSV/TXT files?

Quiz 1: Answer Key & Rationale

1. Horizontal Scaling Unlike traditional databases, Big Data thrives on adding **more nodes** (clusters) rather than bigger, expensive single servers.
2. Veracity This is the "Truth" of data. We address this in the **Silver Layer** by cleaning outliers and nulls.
3. In-Memory Processing Spark keeps data in the **RAM**, avoiding the slow disk-write cycles that limited Hadoop.
4. Bronze Layer It stores data in its **Raw state**. This is essential for legal audits and data lineage in banking.

Databricks Pro-Tip

Unity Catalog allows us to track the **Lineage** of these answers directly back to our original Kaggle CSV!

Bonus Challenge: Governance & Lineage

The "Scenario" Question

A bank auditor asks: *"Why was Customer ID #103 flagged as 'Critical Risk' in your Gold table?"* How does the **Unity Catalog** help you answer this?

The Answer:

- **Data Lineage:** Provides a visual map showing the flow from the `landing_zone` Volume to the `gold_credit_risk` table.
- **Audit Logs:** Records exactly which Spark job and which version of the code calculated that specific risk score.
- **Transition Cleanliness:** It proves that the "Silver Layer" cleaning (imputing interest rates) followed bank policy.

"In Big Data, it's not enough to be right; you must be able to prove HOW you got there."

Advanced AI Tooling: From Code to Insights

Code Generation & Pipeline Automation

- **GitHub Copilot**: Your inline, contextual coding partner. It serves as the primary assistant for writing boilerplate code, autocompleting functions, and streamlining day-to-day scripting directly inside the IDE.
- **Cursor**: A next-generation, AI-first code editor designed for deep codebase navigation, multi-file edits, and context-aware feature implementation.
- **Claude Code**: A powerful agentic interface focused on the execution and automation of complex development workflows, particularly optimized for building and managing the Medallion pipeline.

Advanced AI Tooling: From Code to Insights

Specialized Intelligence & Querying

- **Gemini Genie**: A high-reasoning engine dedicated to Architectural Research and PhD-level logic. This handles complex system design decisions, data modeling theory, and deep logical analysis.
- **Databricks Genie**: A specialized Data Agent that bridges the gap between technical data structures and natural language. It enables direct, conversational querying right on top of the Lakehouse.

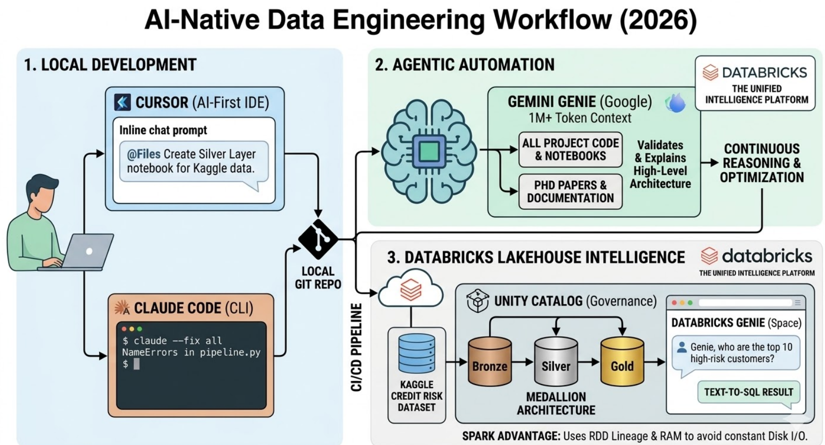
The Shift in Professional Workflow

In 2026, the Data Scientist acts as an **Orchestrator**. We define the governance (Unity Catalog) and the strategy, while AI Agents execute the manual coding and SQL generation.

Advanced AI Tooling: From Code to Insights

Architectural Proposal

Figure: AI-Augmented Data Engineer



WorkShop 1

DataBricks → Genie → Code